



## Primetrade.ai internship assignment

### Trader Performance vs Market Sentiment (Fear & Greed) Analysis

#### Import Libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

#### Load Data

```
In [2]: trades = pd.read_csv("historical_data.csv")
sentiment = pd.read_csv("fear_greed_index.csv")
```

#### Understand the Data

```
In [3]: print(trades.head())
print(trades.info())
print(sentiment.head())
print(sentiment.info())
```

|   | Account                                    | Coin | Execution Price | \ |
|---|--------------------------------------------|------|-----------------|---|
| 0 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9769          |   |
| 1 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9800          |   |
| 2 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9855          |   |
| 3 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9874          |   |
| 4 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9894          |   |

|   | Size Tokens | Size USD | Side | Timestamp IST    | Start Position | Direction | \ |
|---|-------------|----------|------|------------------|----------------|-----------|---|
| 0 | 986.87      | 7872.16  | BUY  | 02-12-2024 22:50 | 0.000000       | Buy       |   |
| 1 | 16.00       | 127.68   | BUY  | 02-12-2024 22:50 | 986.524596     | Buy       |   |
| 2 | 144.09      | 1150.63  | BUY  | 02-12-2024 22:50 | 1002.518996    | Buy       |   |
| 3 | 142.98      | 1142.04  | BUY  | 02-12-2024 22:50 | 1146.558564    | Buy       |   |
| 4 | 8.73        | 69.75    | BUY  | 02-12-2024 22:50 | 1289.488521    | Buy       |   |

|   | Closed PnL | Transaction Hash                                  | Order ID    |
|---|------------|---------------------------------------------------|-------------|
| 0 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 1 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 2 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 3 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 4 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |

|   | Crossed | Fee      | Trade ID     | Timestamp    |
|---|---------|----------|--------------|--------------|
| 0 | True    | 0.345404 | 8.950000e+14 | 1.730000e+12 |
| 1 | True    | 0.005600 | 4.430000e+14 | 1.730000e+12 |
| 2 | True    | 0.050431 | 6.600000e+14 | 1.730000e+12 |
| 3 | True    | 0.050043 | 1.080000e+15 | 1.730000e+12 |
| 4 | True    | 0.003055 | 1.050000e+15 | 1.730000e+12 |

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 211224 entries, 0 to 211223

Data columns (total 16 columns):

| #  | Column           | Non-Null Count  | Dtype   |
|----|------------------|-----------------|---------|
| 0  | Account          | 211224 non-null | object  |
| 1  | Coin             | 211224 non-null | object  |
| 2  | Execution Price  | 211224 non-null | float64 |
| 3  | Size Tokens      | 211224 non-null | float64 |
| 4  | Size USD         | 211224 non-null | float64 |
| 5  | Side             | 211224 non-null | object  |
| 6  | Timestamp IST    | 211224 non-null | object  |
| 7  | Start Position   | 211224 non-null | float64 |
| 8  | Direction        | 211224 non-null | object  |
| 9  | Closed PnL       | 211224 non-null | float64 |
| 10 | Transaction Hash | 211224 non-null | object  |
| 11 | Order ID         | 211224 non-null | int64   |
| 12 | Crossed          | 211224 non-null | bool    |
| 13 | Fee              | 211224 non-null | float64 |
| 14 | Trade ID         | 211224 non-null | float64 |
| 15 | Timestamp        | 211224 non-null | float64 |

dtypes: bool(1), float64(8), int64(1), object(6)

memory usage: 24.4+ MB

None

|   | timestamp  | value | classification | date       |
|---|------------|-------|----------------|------------|
| 0 | 1517463000 | 30    | Fear           | 2018-02-01 |

```

1  1517549400      15  Extreme Fear  2018-02-02
2  1517635800      40           Fear  2018-02-03
3  1517722200      24  Extreme Fear  2018-02-04
4  1517808600      11  Extreme Fear  2018-02-05
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2644 entries, 0 to 2643
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   timestamp             2644 non-null  int64
1   value                 2644 non-null  int64
2   classification        2644 non-null  object
3   date                  2644 non-null  object
dtypes: int64(2), object(2)
memory usage: 82.8+ KB
None

```

### Rows and columns

```
In [4]: print("Trades Shape:", trades.shape)
        print("Sentiment Shape:", sentiment.shape)
```

```

Trades Shape: (211224, 16)
Sentiment Shape: (2644, 4)

```

### Missing values

```
In [5]: print(trades.isnull().sum())
        print(sentiment.isnull().sum())
```

```

Account          0
Coin             0
Execution Price   0
Size Tokens       0
Size USD          0
Side             0
Timestamp IST     0
Start Position    0
Direction         0
Closed PnL        0
Transaction Hash  0
Order ID          0
Crossed           0
Fee              0
Trade ID          0
Timestamp         0
dtype: int64
timestamp        0
value            0
classification    0
date             0
dtype: int64

```

### Clean Data

## Trader Dataset

```
In [32]: trades["Timestamp IST"] = pd.to_datetime(
        trades["Timestamp IST"],
        format="%d-%m-%Y %H:%M"
    )
trades["Date"] = trades["Timestamp IST"].dt.date
```

## Sentiment Dataset

```
In [33]: sentiment = sentiment.rename(columns={
        "date": "Date",
        "classification": "Classification"
    })
sentiment["Date"] = pd.to_datetime(sentiment["Date"]).dt.date
```

```
In [34]: print(trades["Date"].head())
print(sentiment["Date"].head())
```

```
0    2024-12-02
1    2024-12-02
2    2024-12-02
3    2024-12-02
4    2024-12-02
Name: Date, dtype: object
0    2018-02-01
1    2018-02-02
2    2018-02-03
3    2018-02-04
4    2018-02-05
Name: Date, dtype: object
```

## Merge

```
In [35]: merged = pd.merge(
        trades,
        sentiment,
        on="Date",
        how="left"
    )
```

```
In [36]: print(merged.head())
print(merged["Classification"].value_counts(dropna=False))
```

|   | Account                                    | Coin | Execution Price \ |
|---|--------------------------------------------|------|-------------------|
| 0 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9769            |
| 1 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9800            |
| 2 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9855            |
| 3 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9874            |
| 4 | 0xae5eacaf9c6b9111fd53034a602c192a04e082ed | @107 | 7.9894            |

|   | Size Tokens | Size USD | Side | Timestamp IST       | Start Position | Direction \ |
|---|-------------|----------|------|---------------------|----------------|-------------|
| 0 | 986.87      | 7872.16  | BUY  | 2024-12-02 22:50:00 | 0.000000       | Buy         |
| 1 | 16.00       | 127.68   | BUY  | 2024-12-02 22:50:00 | 986.524596     | Buy         |
| 2 | 144.09      | 1150.63  | BUY  | 2024-12-02 22:50:00 | 1002.518996    | Buy         |
| 3 | 142.98      | 1142.04  | BUY  | 2024-12-02 22:50:00 | 1146.558564    | Buy         |
| 4 | 8.73        | 69.75    | BUY  | 2024-12-02 22:50:00 | 1289.488521    | Buy         |

|   | Closed PnL | Transaction Hash                                  | Order ID    |
|---|------------|---------------------------------------------------|-------------|
| 0 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 1 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 2 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 3 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |
| 4 | 0.0        | 0xec09451986a1874e3a980418412fcd0201f500c95bac... | 52017706630 |

|   | Crossed | Fee      | Trade ID     | Timestamp    | Date       | timestamp \  |
|---|---------|----------|--------------|--------------|------------|--------------|
| 0 | True    | 0.345404 | 8.950000e+14 | 1.730000e+12 | 2024-12-02 | 1.733117e+09 |
| 1 | True    | 0.005600 | 4.430000e+14 | 1.730000e+12 | 2024-12-02 | 1.733117e+09 |
| 2 | True    | 0.050431 | 6.600000e+14 | 1.730000e+12 | 2024-12-02 | 1.733117e+09 |
| 3 | True    | 0.050043 | 1.080000e+15 | 1.730000e+12 | 2024-12-02 | 1.733117e+09 |
| 4 | True    | 0.003055 | 1.050000e+15 | 1.730000e+12 | 2024-12-02 | 1.733117e+09 |

|   | value | Classification |
|---|-------|----------------|
| 0 | 80.0  | Extreme Greed  |
| 1 | 80.0  | Extreme Greed  |
| 2 | 80.0  | Extreme Greed  |
| 3 | 80.0  | Extreme Greed  |
| 4 | 80.0  | Extreme Greed  |

|               | Classification |
|---------------|----------------|
| Fear          | 61837          |
| Greed         | 50303          |
| Extreme Greed | 39992          |
| Neutral       | 37686          |
| Extreme Fear  | 21400          |
| NaN           | 6              |

Name: count, dtype: int64

## Winning Trade

```
In [37]: merged["Win"] = merged["Closed PnL"] > 0
```

## Daily PnL

```
In [38]: daily_pnl = merged.groupby(
["Account", "Date"]
)["Closed PnL"].sum().reset_index()
```

## Win Rate

```
In [39]: win_rate = merged.groupby("Account")["Win"].mean()*100
```

## Average Trade Size

```
In [40]: avg_trade = merged.groupby("Account")["Size USD"].mean()
```

## Number of Trades

```
In [41]: trade_frequency = merged.groupby("Date").size()
```

## Long Short Ratio

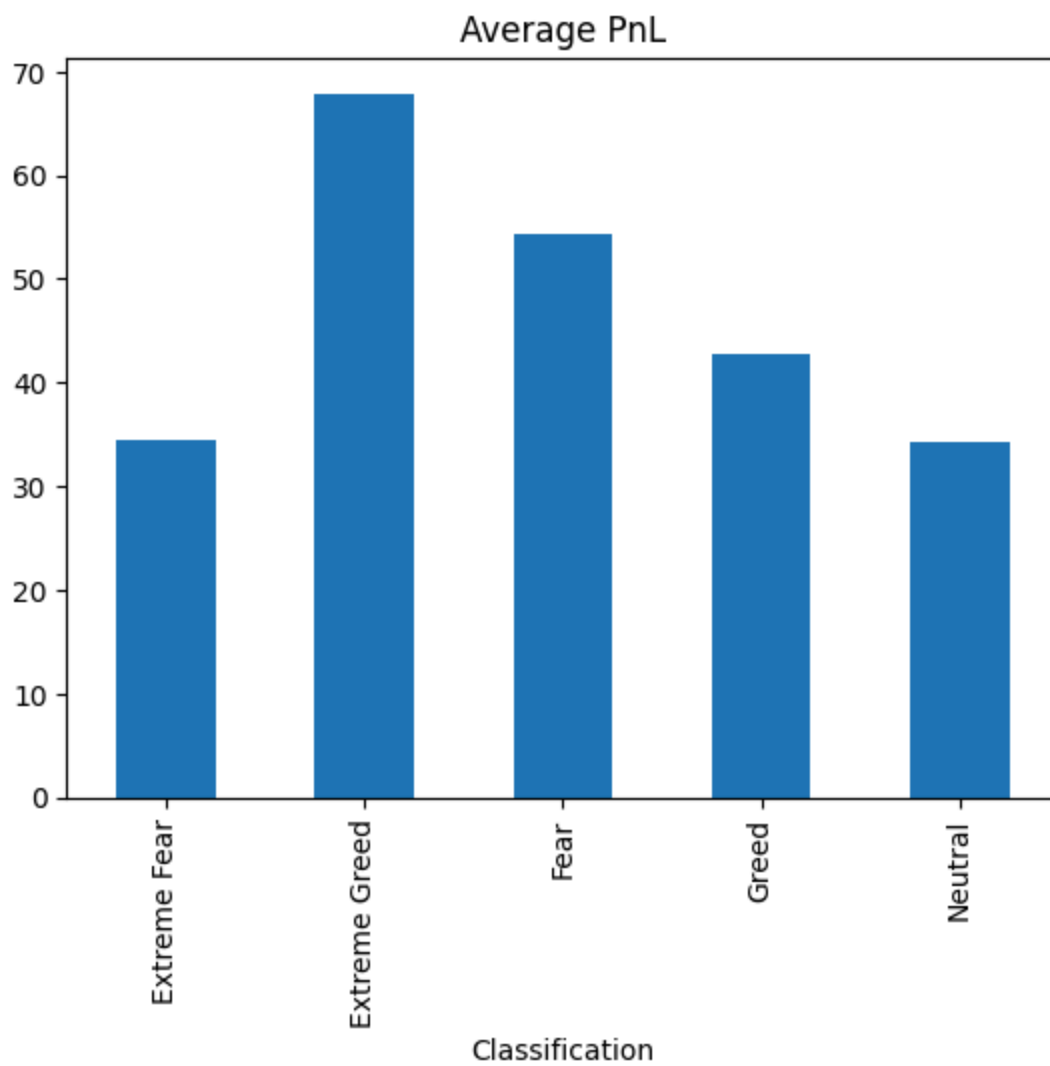
```
In [42]: print(merged["Direction"].value_counts())
```

```
Direction
Open Long          49895
Close Long         48678
Open Short         39741
Close Short        36013
Sell               19902
Buy               16716
Spot Dust Conversion    142
Short > Long          70
Long > Short          57
Auto-Deleveraging      8
Liquidated Isolated Short  1
Settlement            1
Name: count, dtype: int64
```

## Exploratory Data Analysis

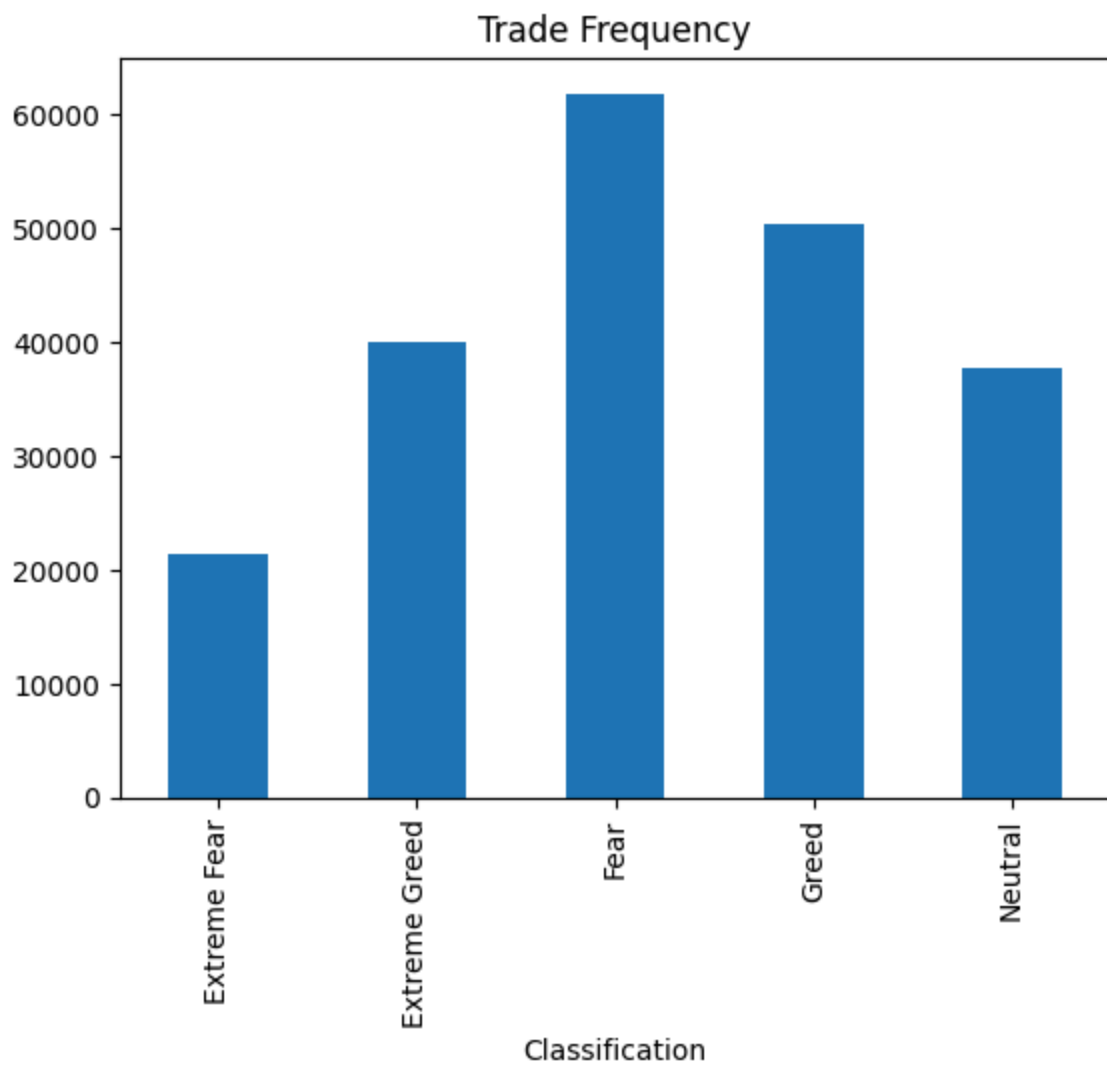
### Average PnL by Sentiment

```
In [43]: merged.groupby(
    "Classification"
)["Closed PnL"].mean().plot(kind="bar")
plt.title("Average PnL")
plt.show()
```



### Number of Trades

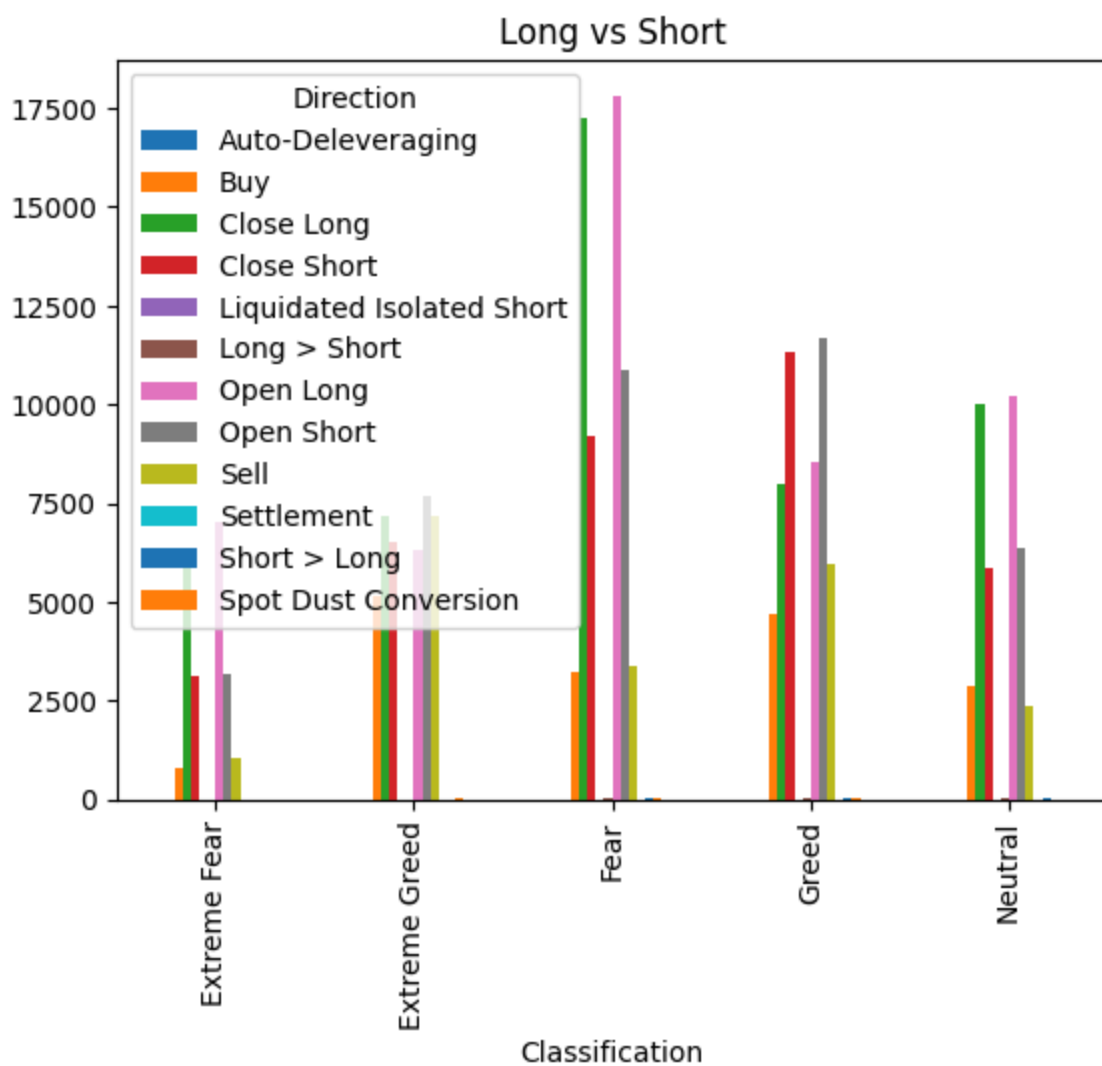
```
In [45]: merged.groupby(  
    "Classification"  
).size().plot(kind="bar")  
plt.title("Trade Frequency")  
plt.show()
```



### Long vs Short

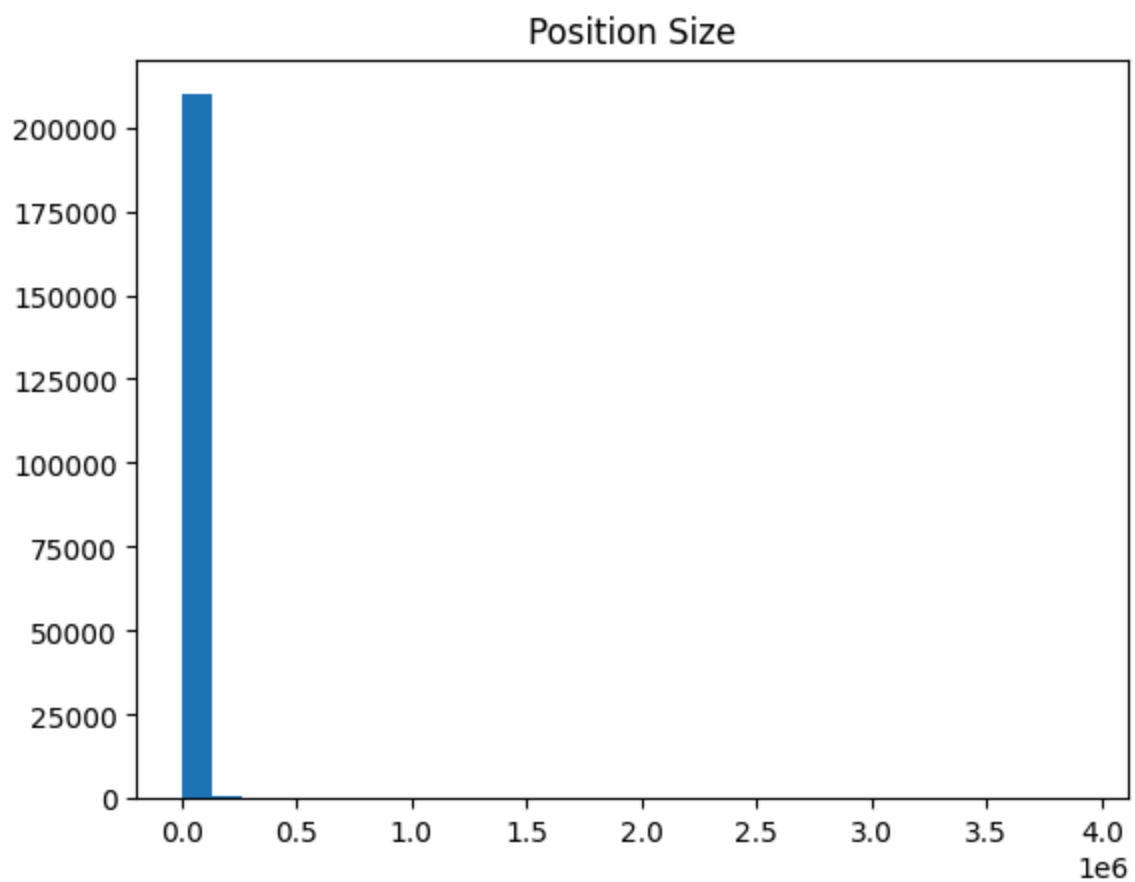
```
In [46]: pd.crosstab(  
merged["Classification"],  
merged["Direction"]  
) .plot(kind="bar")  
plt.title("Long vs Short")  
plt.show()
```





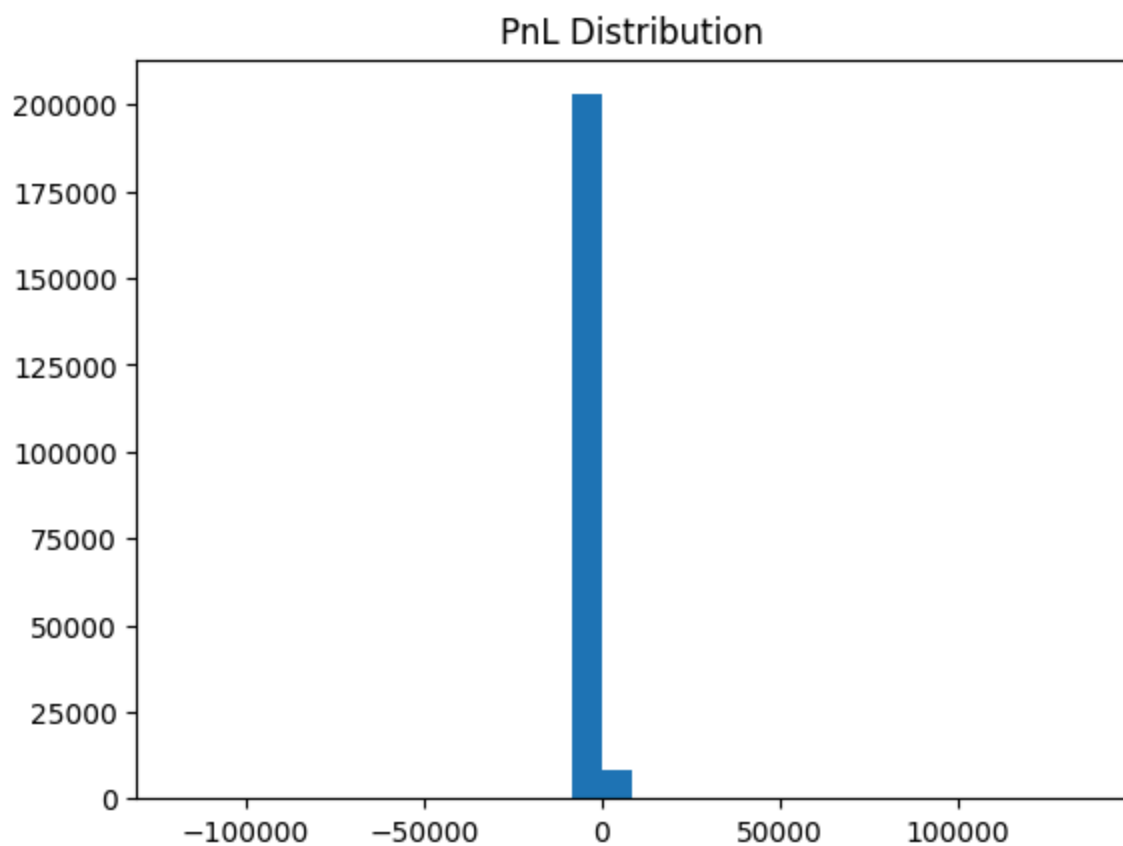
### Position Size Distribution

```
In [47]: plt.hist(
merged["Size USD"],
bins=30
)
plt.title("Position Size")
plt.show()
```



### Closed PnL Distribution

```
In [48]: plt.hist(  
merged["Closed PnL"],  
bins=30  
)  
plt.title("PnL Distribution")  
plt.show()
```



### Does performance differ?

```
In [49]: merged.groupby(
          "Classification"
        )[
          "Closed PnL"
        ].agg(
          ["mean", "median", "sum"]
        )
```

```
Out[49]:
```

|                       | mean      | median | sum          |
|-----------------------|-----------|--------|--------------|
| <b>Classification</b> |           |        |              |
| <b>Extreme Fear</b>   | 34.537862 | 0.0    | 7.391102e+05 |
| <b>Extreme Greed</b>  | 67.892861 | 0.0    | 2.715171e+06 |
| <b>Fear</b>           | 54.290400 | 0.0    | 3.357155e+06 |
| <b>Greed</b>          | 42.743559 | 0.0    | 2.150129e+06 |
| <b>Neutral</b>        | 34.307718 | 0.0    | 1.292921e+06 |

### Trade frequency

```
In [50]: merged.groupby(
```

```
"Classification"  
) .size()
```

```
Out[50]: Classification  
Extreme Fear      21400  
Extreme Greed     39992  
Fear              61837  
Greed             50303  
Neutral           37686  
dtype: int64
```

### Average Position Size

```
In [52]: merged.groupby(  
"Classification"  
)["Size USD"].mean()
```

```
Out[52]: Classification  
Extreme Fear      5349.731843  
Extreme Greed     3112.251565  
Fear              7816.109931  
Greed             5736.884375  
Neutral           4782.732661  
Name: Size USD, dtype: float64
```

### Win Rate

```
In [53]: merged.groupby(  
"Classification"  
)["Win"].mean()*100
```

```
Out[53]: Classification  
Extreme Fear      37.060748  
Extreme Greed     46.494299  
Fear              42.076750  
Greed             38.482794  
Neutral           39.699093  
Name: Win, dtype: float64
```

### Trader Segments

#### Frequent Traders

```
In [54]: segment1 = merged.groupby(  
"Account"  
) .size().sort_values(  
ascending=False  
)
```

#### Highest Profit

```
In [55]: segment2 = merged.groupby(  
"Account"
```

```
"Account"  
)["Closed PnL"].sum().sort_values(  
ascending=False  
)
```

### Biggest Position Size

```
In [28]: segment3 = merged.groupby(  
"Account"  
)["Size USD"].mean().sort_values(  
ascending=False  
)
```